Molecule – Flavour: answer key

#	Molecule	Flavour card	Found in	Receptor	Fun fact
1	**L-Glutamate** (2S)-2-Aminopentanedioic acid	**Umami**	Parmesan · tomato · kombu	TAS1R1 / TAS1R3	Identified as the taste of kombu dashi by Kikunae Ikeda in 1908.
2	**Sucrose** β-D-Fructofuranosyl α-D-glucopyranoside	**Sweet**	Sugar cane · sugar beet · sweets	TAS1R2 / TAS1R3	Cats carry a broken Tas1r2 gene and cannot taste sweet.
3	**Citric acid** 2-Hydroxypropane-1,2,3-tricarboxylic acid	**Sour**	Lemon · lime	OTOP1 (proton channel)	The sour receptor was only identified in 2018–2019.
4	**Sodium chloride**	**Salty**	Sea salt · crisps	ENaC	The diuretic amiloride blocks ENaC and dulls part of salt taste.
5	**Quinine** (8S,9R)-6'-Methoxycinchonan-9-ol	**Bitter**	Tonic water	Several TAS2Rs	Fluoresces blue under UV light – try it with a G&T.
6	**Caffeine** 1,3,7-Trimethylxanthine	**Bitter**	Coffee · tea	TAS2R7/10/14/43/46	Its bitterness is detected by at least five different TAS2Rs.
7	**Phenylthiocarbamide (PTC)**	**Bitter… or nothing?**	Broccoli · Brussels sprouts	TAS2R38	About a quarter of people can't taste it; TAS2R38 haplotype (PAV/AVI) predicts who.
8	**Glycyrrhizin**	**Intensely sweet**	Liquorice root · salmiak	TAS1R2 / TAS1R3	30–50× sweeter than sucrose, with a slow, lingering onset.
9	**Capsaicin** (6E)-N-(4-Hydroxy-3-methoxybenzyl)-8-methylnon-6-enamide	**Burning heat**	Chilli peppers	TRPV1	Cloning TRPV1 with capsaicin earned David Julius the 2021 Nobel Prize.
10	**(−)-Menthol** (1R,2S,5R)-2-Isopropyl-5-methylcyclohexan-1-ol	**Cooling**	Peppermint · cough drops	TRPM8	TRPM8 is the cold sensor itself – menthol just lowers its activation threshold.
11	**Allyl isothiocyanate** 3-Isothiocyanatoprop-1-ene	**Sharp, up-the-nose**	Wasabi · mustard · horseradish	TRPA1	Volatile, so it hits the nose rather than lingering on the tongue.
12	**Cinnamaldehyde** (2E)-3-Phenylprop-2-enal	**Warm & spicy**	Cinnamon · kanelbullar	TRPA1 (+ olfactory receptors)	An electrophile: it covalently modifies cysteines on TRPA1.
13	**Hydroxy-α-sanshool** (2E,6Z,8E,10E)-N-(2-Hydroxy-2-methylpropyl)dodeca-2,6,8,10-tetraenamide	**Tingling & numbing**	Sichuan pepper	KCNK3/9/18 (two-pore K+ channels)	The tingle feels like ~50 Hz vibration on the lips.
14	**Piperine** (2E,4E)-5-(1,3-Benzodioxol-5-yl)-1-(piperidin-1-yl)penta-2,4-dien-1-one	**Peppery bite**	Black pepper · cacio e pepe	TRPV1 / TRPA1	Same receptor as capsaicin, but a much gentler burn.
15	**(R)-(−)-Carvone**	**Spearmint**	Spearmint · chewing gum	Olfactory receptors	Mirror image of (S)-carvone: same atoms, completely different smell.

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16	**(S)-(+)-Carvone**	**Caraway & dill**	Dill · caraway · aquavit	Olfactory receptors	Mirror image of (R)-carvone, and the nose can tell them apart.
17	**Vanillin** 4-Hydroxy-3-methoxybenzaldehyde	**Vanilla**	Vanilla pods · ice cream	OR10G4	Most vanillin is synthetic, made from guaiacol or lignin rather than orchids.
18	**Androstenone**	**Urine? Sandalwood? Nothing?**	Boar meat ('boar taint')	OR7D4	OR7D4 genotype predicts whether it smells foul, sweet, or of nothing at all.
19	**(E)-2-Decenal**	**Fresh herb… or soap?**	Coriander leaves	OR6A2 locus	A variant near OR6A2 is linked to finding coriander soapy.
20	**Isoamyl acetate** 3-Methylbutyl acetate	**Banana / pear drops**	Bananas · pear-drop sweets	Olfactory receptors	Also a honeybee alarm pheromone.